



Product Requirements Document

Smart AC Intelligence Platform · monitoring, energy saving, maintenance, payment-linked control and carbon

Document	D6 — Product Requirements Document (PRD)
Product	ARUSIQ Smart AC Intelligence Platform
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Relationship to other documents	Consolidates and supersedes section 2 (PRD) of the Smart AC Phase 1 Product Definition Pack v0.1. Epics EP-01..EP-15 are kept; EP-16..EP-22 are added. Detailed by D1 Use Cases, D2 UI/UX, D3 Architecture, D4 Rules & Carbon Methodology; scheduled by the Stage 1 WBS and the Team & Budget Plan.

STATUS Draft for stakeholder validation. This document is part of the Phase 1 design set and extends the Smart AC Phase 1 Product Definition Pack (URS, PRD, roles, journeys, sitemap, acceptance criteria). Identifiers UR-, EP-, J0x, SCR- and AC- refer to that pack.

Contents

- 1. Product vision3
- 2. Problem and opportunity3
- 3. Goals and success metrics3
- 4. Personas and jobs to be done3
- 5. Scope by phase4
- 6. Feature epics 4
- 7. Functional requirements5
 - 7.1 Identity, language and voice (EP-01, EP-02, EP-13)5
 - 7.2 Dashboards and assets (EP-02, EP-03, EP-04) 6
 - 7.3 Signals, alerts and predictive maintenance (EP-05, EP-06, EP-16)6
 - 7.4 Maintenance workflow (EP-07) 7
 - 7.5 Remote control and energy automation (EP-08, EP-17) 7
 - 7.6 Payment-linked service management (EP-09) 7
 - 7.7 Energy, air quality and comfort (EP-10, EP-11, EP-18)8
 - 7.8 Carbon, Digital MRV and platform integration (EP-12, EP-19) 8
 - 7.9 Gadget, firmware and device trust (EP-14, EP-20)9
 - 7.10 HVAC (EP-21)**Error! Bookmark not defined.**
 - 7.11 Administration and platform (EP-15, EP-22) 9
- 8. Non-functional requirements 9
- 9. UX principles and design references9
- 10. Analytical integrity and safety policy10
- 11. Dependencies, assumptions and risks10
- 12. Release plan and acceptance10
- 13. Open questions 11
- 14. Traceability summary11

1. Product vision

Every air conditioner becomes a monitored, controllable and accountable asset. A small, affordable gadget hidden inside the unit, working with any brand, tells the owner, the technician and headquarters what is happening in every part of the machine, what it costs, how much carbon it saves, and when it will need service, before it breaks. The same platform lets the company run rent-to-own and maintenance services responsibly, and turns measured energy savings into audit-ready carbon evidence that can be reported, exchanged or offset.

ONE-LINE PITCH Monitor every aspect of the air conditioner, fix problems before they happen, cut energy bills by 10–20 % or more, and prove the carbon saved.

2. Problem and opportunity

- **Reactive service.** Occupants discover problems after comfort is lost; technicians arrive without diagnostic history; emergency repairs cost more than timely maintenance.
- **No visibility of cost and waste.** Units run in empty rooms, at 18–22 °C setpoints and through peak tariffs; nobody sees the bill until it arrives.
- **Commercial risk in rent-to-own.** The company has no proportionate way to respond when payments stop, and no evidence trail if it does.
- **Carbon claims without proof.** Energy-efficiency savings are estimated, not metered, so they cannot become sustainability reports or credits.
- **Opportunity.** Millions of split units in tropical markets; a brand-agnostic retrofit gadget plus a service platform serves homes, offices and, in Phase 2, HVAC.

3. Goals and success metrics

Goal	Metric	Target	Measured how
G1 Understand status in one minute	Time for a new user to answer 'what needs attention'	< 60 s	User test, Stage 1
G2 Proactive maintenance	Share of red alerts confirmed by technician visit	≥ 70 % (pilot)	Work-order verdicts
G3 Measurable savings	Energy saved vs baseline on pilot units	10–20 % or more	Metered kWh vs baseline (D4 §5)
G4 Responsible commercial control	Restrictions applied without a warning, grace period and dual approval	0	Audit log
G5 Sustainability readiness	Periods with complete, signed MRV packages	≥ 90 % completeness	MRV service
G6 Affordability and scale	Gadget bill of materials at 1,000 units; brands supported by codebook	< USD 12 base; top 10 brands	Hardware costing; codebook coverage
G7 Delight	Client app rating and monthly active use	≥ 4.5 / 5; ≥ 60 % MAU	Store and analytics

4. Personas and jobs to be done

Persona	Primary outcome	Core jobs	Dashboard emphasis
Client / End user (home or office)	Comfort and clean air at the lowest bill; no surprises	Check status, respond to alerts, control and schedule, request service, pay, see carbon saved	Strong visual: comfort now, energy today, savings, alerts, maintenance, carbon hero

Persona	Primary outcome	Core jobs	Dashboard emphasis
Internal technician	Fix it right the first time	Prioritise, review evidence, inspect per part, document, close	Work queue, priority alerts, component diagram, device health
Third-party technician	Complete assigned jobs securely	Same as internal within assigned scope	Assigned work only
Admin / HQ	Run fleet, service and commercial control	Monitor, dispatch, approve restrictions, configure rules, report carbon	Fleet map, critical events, service funnel, payment attention, command activity, energy portfolio, MRV status
Business owner	Grow services and markets	Rent-to-own, maintenance, carbon-as-a-service, HVAC expansion	Executive KPIs and reports

5. Scope by phase

Phase	Scope	Out of scope	Exit outcome
1A Clickable prototype (2-week sprint)	Sign in/out/forgot password; language switch EN/ID; text assistant with voice transcript; three role dashboards; Home/Office hierarchy; unit detail with simulated live data; red/orange/green alerts; maintenance and service requests; payment notices and safe-restriction simulation; energy and savings vs normal; Scope 2 estimate; MRV and carbon-platform mock; tamper simulation	Real devices, real notifications, real payments	Pitch-ready prototype; Stage 2 go/no-go
1B Functional platform	Authentication with MFA, multi-tenant database, telemetry ingestion, rules engine, notifications (WhatsApp, email, push), work orders, payments and policy engine, schedules and automations, baseline learning, Scope 2 reports	Registry-grade carbon credits	Platform ready for devices
1C Gadget and firmware	Small hidden device: sensors per part group, smart-thermostat control by IR or relay, tamper detection, signed metering, OTA; 30-unit pilot; SDPPI application	Custom cost-optimised board (follows pilot)	Live pilot with measured savings
2 HVAC	Central, cassette, VRF through BACnet/Modbus and vendor gateways; ventilation control; site-level MRV	—	Same product for large buildings
Future	Verified credits under an accepted methodology; tokenized micro-offsets on mainnet; carbon trading API live; solar and battery integration	—	Carbon-as-a-service at scale

6. Feature epics

Epics EP-01 to EP-15 come from the Product Definition Pack; EP-16 to EP-22 are added from the discussion. Priority is for the phase named.

Epic	Name	Behaviour	Priority · phase
EP-01	Identity and demo access	Sign in, sign out, forgot password, seeded role accounts; MFA later	Must · 1A
EP-02	Role-based experience	Distinct navigation and strong graphical dashboards for Client, Technician (internal, 3rd party) and Admin/HQ	Must · 1A
EP-03	Asset explorer	Home or Office → property → floor/area → room/space → unit; status roll-up	Must · 1A
EP-04	Live operations view	Status, temperature, humidity, power, runtime, connectivity per unit	Must · 1A

Epic	Name	Behaviour	Priority · phase
EP-05	Health and diagnostics	Part-level health for indoor, outdoor and electrical groups; suspected fault, evidence, confidence, recommendation	Must · 1A
EP-06	Alerts and notification centre	Green/orange/red/grey; in-app plus WhatsApp and email with delivery status	Must · 1A
EP-07	Maintenance workflow	Service request, work order, checklist per part, history, closure, post-service check; regular, reactive, preventive	Must · 1A
EP-08	Remote control and schedules	Confirmed commands with Sent → Acknowledged → Verified; weekly schedules	Must · 1A
EP-10	Energy intelligence	Real-time usage, tariff cost, baseline vs actual, savings with the solution vs normal	Must · 1A
EP-11	Indoor environment	Temperature, humidity, CO ₂ , dust; air-quality interpretation; filter alerts	Should · 1A
EP-12	Carbon and Digital MRV preview	Scope 2 estimate, avoided emissions, audit-readiness, mock carbon platform	Should · 1A
EP-13	Multilingual and voice	Any language via locale packs; text and voice assistant with transcript and confirmation	Should · 1A
EP-14	Device trust	Heartbeat, offline, power loss, possible-tamper and removal events	Must · 1A
EP-15	Administration	Users, roles, thresholds, tariffs, emission factors, audit	Should · 1A
EP-16	Predictive maintenance engine	Rules and anomaly models per part (vibration, temperature, refrigerant, filter, current); health scores; time-to-failure band 1A: Mock if possible and applicable	Must · 1B
EP-17	Energy automation	Occupancy auto-off, geofencing with consent, time-of-use pre-cooling, weather-linked automated comfort, learned routine; solar and battery link later 1A: Mock if possible and applicable	Should · 1B
EP-18	Ventilation and health protection	Stale-air alerts, ventilation triggers where a fan or damper exists, humidity and dust bands 1A: Mock if possible and applicable	Should · 1B / 2
EP-19	Carbon-as-a-service	Signed metering, baseline method, MRV packages, verifier workflow, Scope 2 reports, offsets and tokenization via platform API 1A: Mock if possible and applicable	Should · 1C / Future
EP-22	Platform, security and integrations	Multi-tenant cloud, identity, MQTT device gateway, notification, payment and carbon adapters, public API and webhooks	Must · 1B

7. Functional requirements

Functional requirements state what the product does for each epic. Each maps to the user requirements in D5 and is detailed in the use cases (D1) and design documents (D2 to D4).

7.1 Identity, language and voice (EP-01, EP-02, EP-13)

FR ID	Functional requirement	Priority	Satisfies (URS)	Phase
FR-01	Provide Sign In with email or phone and password, lockout after 5 failures, and role-based landing.	Must	UR-AUTH-01, 04	1A
FR-02	Provide Sign Out for this device and for all devices.	Must	UR-AUTH-02	1A
FR-03	Provide Forgot Password with neutral confirmation, time-limited link or code by email or WhatsApp, and password policy.	Must	UR-AUTH-03	1A

FR ID	Functional requirement	Priority	Satisfies (URS)	Phase
FR-04	Provide a language switch on every screen; all strings, templates and formats from locale packs; EN and ID at launch.	Must	UR-LANG-01, 02	1A

7.2 Dashboards and assets (EP-02, EP-03, EP-04)

FR ID	Functional requirement	Priority	Satisfies (URS)	Phase
FR-10	Provide a Client overview with hero (carbon reduced this month), KPI tiles, room rows with comfort verdict, active alerts, energy sparkline and maintenance timeline.	Must	UR-CLI-01	1A
FR-11	Provide a Technician overview with work queue, priority alerts, route and device health.	Must	UR-TEC-01	1A
FR-12	Provide an Admin/HQ overview with fleet map, critical events, service funnel, payment attention, command activity, energy portfolio and MRV status.	Must	UR-ADM-01	1A
FR-13	Provide the category and space hierarchy (Home or Office → property → floor → room → unit) with worst-status roll-up, search and, for Admin, add and map view.	Must	UR-ASSET-01, UR-ADM-02	1A
FR-14	Provide a unit detail page with current values and freshness, trends, part-level health, controls, carbon card, schedule and history.	Must	UR-CLI-02, 03	1A
FR-15	Show grey values with last-seen time when data is stale; never fabricate values.	Must	UR-SYS-02	1A

7.3 Signals, alerts and predictive maintenance (EP-05, EP-06, EP-16)

FR ID	Functional requirement	Priority	Satisfies (URS)	Phase
FR-20	Monitor and classify each part in the indoor group (air filter, evaporator coil, blower motor and fan, condensate drain line and pan, vents and louvers), outdoor group (condenser coil, compressor, condenser fan and blades, refrigerant lines) and electrical group (thermostat and sensors, capacitor and contactor, wiring) as green, orange, red or grey using the rule catalogue in D4.	Must	UR-MNT-01..04	1A sim / 1B
FR-21	Generate alerts with evidence (signals, thresholds, duration), confidence, likely cause, impact if ignored and recommended action.	Must	UR-MNT-01, UR-TEC-04	1A
FR-22	Deliver alerts in-app and by push, WhatsApp and email per severity and preference, in the user's language, with delivery status and escalation timers.	Must	UR-CLI-08, UR-COM-01, 02	1A
FR-25	Label predictions suspected until a technician verdict; run a post-service check and reopen the work order if readings do not improve.	Must	UR-MNT-08	1A / 1B
FR-26	Notify when cooling load is double the expected load (open window, uninsulated wall).	Should	UR-MNT-09	1B
FR-27	Let Admin configure thresholds, hysteresis, durations and escalation with a preview of affected units, versioned and audited.	Should	UR-ADM-09	1A preview / 1B

7.4 Maintenance workflow (EP-07)

FR ID	Functional requirement	Priority	Satisfies (URS)	Phase
FR-30	Let a client create a service request prefilled from an alert or unit, with type (regular, reactive, preventive), photos, preferred slots and channel, and track it.	Must	UR-CLI-09	1A
FR-31	Let Admin create, prioritise and assign work orders with SLA; auto-create preventive work orders on contract cadence or runtime.	Must	UR-ADM-08, UR-MNT-07	1A
FR-32	Let a technician review evidence, complete the part-group checklist with measurements and photos, record parts and findings, close or escalate.	Must	UR-TEC-03..08	1A
FR-33	Show maintenance history, contracts, warranty dates, renewals and upcoming visits; export service reports.	Should	UR-CLI-10	1A
FR-34	Scope third-party technicians to assigned work orders only.	Must	UR-TEC-02	1A

7.5 Remote control and energy automation (EP-08, EP-17)

1A: Simulated/Mocked if possible and applicable

FR ID	Functional requirement	Priority	Satisfies (URS)	Phase
FR-40	Let authorised users change power, mode (Cool, Fan, Eco, Off), setpoint and fan speed with a confirmation that shows target, current value and any policy limit; show Sent → Acknowledged → Verified.	Must	UR-CLI-07, UR-ADM-07	1A sim / 1C
FR-41	Provide weekly schedules (off when nobody is home, on before arrival) with offline fallback.	Must	UR-ENE-02	1A
FR-42	Provide occupancy auto-off using room presence.	Should	UR-ENE-03	1B
FR-43	Provide geofencing with explicit consent: throttle cooling when the household leaves, restore on approach; store zone events only.	Should	UR-ENE-03, UR-CLI-12	1B
FR-44	Provide time-of-use shift: pre-cool when tariff is low, rest during peak windows; show cost effect.	Should	UR-ENE-04	1B
FR-45	Provide automated comfort: learned routine, weather-linked setpoint, automatic Dry or ventilation, with every action logged.	Should	UR-ENE-06	1B
FR-47	Provide group commands for a floor or property with per-unit results.	Should	UR-ADM-07	1B

7.6 Payment-linked service management (EP-09)

FR ID	Functional requirement	Priority	Satisfies (URS)	Phase
FR-50	Show clients their balance, due date, payment methods (card, debit, virtual account, e-wallet), payment instructions, notice history and any restriction state; let them pay through the gateway. 1A: Simulated/Mocked if possible and applicable	Must	UR-CLI-11	1A sim / 1B
FR-51	Send reminders and overdue notices by WhatsApp and email with a payment link, in the client's language, and record delivery. 1A: mock if possible and applicable	Must	UR-PAY-02	1A sim / 1B

FR ID	Functional requirement	Priority	Satisfies (URS)	Phase
FR-52	Implement the restriction ladder: reminder → minimum setpoint raised (for example 26 °C) → Eco lock and limited hours → off; each step with notice period, grace period, requester plus approver, and audit. 1a: simulated/mocked if possible and applicable	Must	UR-PAY-01, 03, UR-ADM-06	1A sim / 1B
FR-53	Block the off step for spaces flagged health-sensitive; allow fan-only in hot spells by policy.	Must	UR-PAY-03	1B
FR-54	Restore normal operation automatically on payment confirmation and verify at the device.	Must	UR-PAY-04	1B
FR-55	Expose the ladder as a policy so other programmes (contract expiry, demand response, landlord rules) can reuse it.	Should	UR-PAY-05	1B
FR-56	Enforce restrictions on the device locally until a signed release arrives, even if the cloud is unreachable.	Must	UR-PAY-01, UR-DEV-06	1C

7.7 Energy, air quality and comfort (EP-10, EP-11, EP-18)

FR ID	Functional requirement	Priority	Satisfies (URS)	Phase
FR-60	Meter each unit's power and energy in real time; show watts, kWh today and month, and cost from the tariff.	Must	UR-ENE-01	1A sim / 1C
FR-61	Show normal (adjusted baseline) versus with-solution consumption, the saving in kWh, cost and percentage, and the method and completeness behind it.	Must	UR-ENE-07, UR-CLI-05	1A
FR-62	Attribute savings to levers (Eco, setpoint ≥ 23 °C, schedule, occupancy, filter cleaning) and show 'how you save' actions.	Should	UR-ENE-07	1A
FR-63	Measure and display CO ₂ ppm, humidity, PM2.5 and TVOC where fitted with plain-language bands; state which sensors are missing. 1A: Mock if possible/applicable	Should	UR-AIR-01	1A sim / 1C
FR-64	Raise stale-air alerts on high CO ₂ with a ventilation recommendation; trigger a connected fan, fresh-air unit or HVAC damper when available; never claim fresh-air control for a split unit alone. 1A: Mock if possible/applicable	Should	UR-AIR-02, 03	1B / 2
FR-65	Alert when filters need washing or replacing from filter-health rules and run hours.	Must	UR-AIR-04	1A
FR-66	Keep indoor CO ₂ ppm and kgCO ₂ e emissions on separate cards with distinct labels.	Must	UR-AIR-06	1A

7.8 Carbon, Digital MRV and platform integration (EP-12, EP-19)

FR ID	Functional requirement	Priority	Satisfies (URS)	Phase
FR-70	Calculate Scope 2 emissions (location- and market-based) as kWh × versioned regional grid factor with source and effective date shown.	Must	UR-CAR-01	1A
FR-71	Calculate avoided emissions from saved kWh × grid factor; label estimated, provisional (completeness < 90 %) or verified. 1A: mocked/simulated if possible and applicable	Must	UR-CAR-02, 08	1A
FR-73	Produce corporate sustainability reports by tenant, floor, site and period as PDF, CSV and API. 1A: Simple Report only	Should	UR-CAR-06	1B

FR ID	Functional requirement	Priority	Satisfies (URS)	Phase
FR-74	Connect to carbon management platforms and registries through a provider-neutral adapter; let owners retire, transfer or mint verified reductions as micro-offsets with dual approval; store certificates and token IDs. 1A: Mocked/simulated if possible and applicable	Should	UR-CAR-04, 05	Future (mock 1A)

7.9 Gadget, firmware and device trust (EP-14, EP-20)

FR ID	Functional requirement	Priority	Satisfies (URS)	Phase
FR-83	Detect enclosure opening, movement or shock, cable disconnect and power loss (last-gasp), send heartbeats, and raise possible-tamper and removal alerts; maintenance mode suppresses alerts during planned service. 1A: Mocked/simulated if possible and applicable	Must	UR-DEV-04, 05	1A sim / 1C

7.11 Administration and platform (EP-15, EP-22)

FR ID	Functional requirement	Priority	Satisfies (URS)	Phase
FR-100	Manage organisations, users, roles and scopes; invite in the user's language.	Should	UR-ADM-04	1A
FR-101	Manage tariffs and grid emission factors as versioned tables with effective dates.	Should	UR-ADM-10	1A
FR-102	Keep an append-only audit of sensitive actions, approvals, deliveries, commands and calculations; export.	Must	UR-ADM-13	1A

8. Non-functional requirements

Area	Requirement	Target	Phase
Real time	Unit change visible in the app; critical alert delivered	≤ 30 s; ≤ 60 s	1B
Availability	Cloud platform uptime; local policy enforcement when offline	99.5 %; 72 h buffer	1C
Security	TLS everywhere, per-device certificates, MFA for admins, encrypted storage, no card data on platform	Pen test passed before launch	1B
Privacy	Consent for voice and location; presence only; retention schedule; data-protection law per market	Documented and enforced	1B
Accessibility	Status never colour-only; keyboard and screen reader; voice with text alternative; contrast AA	Lighthouse a11y ≥ 90	1A
Responsiveness	Desktop, tablet, mobile; installable PWA; store apps for geofencing	375–1536 px without horizontal scroll	1A
Localisation	Locale packs for UI, templates, formats; tariff and grid factor per region	New language without code change	1A
Scalability	Units per organisation; message throughput	10,000 units; 1 msg/s/unit peak	1B
Data integrity	Simulated, estimated, provisional, verified labels ;	100 % of figures labelled	1A

9. UX principles and design references

- Status first: colour and shape answer the question before the number (D2 §1).

- Strong graphical dashboards for all three roles, sharing one design system and one asset hierarchy.
- Evidence before prediction; safe control with confirmation and verification; transparent numbers with disclosure.
- Any language, any input: locale packs and an assistant that accepts text or voice with a transcript.
- Reference layout: sidebar on desktop, drawer and bottom tabs on mobile, dark hero, KPI tiles, per-unit cards, activity list (build brief CLAUDE_CODE_INSTRUCTION_ArusIQ_App.md).

10. Analytical integrity and safety policy

SAFE CONTROL The product must not suggest immediate automatic shutdown for missed payment. The demonstrated and implemented path is: overdue notice → grace period → administrator review → user warning → approved restricted-service mode (warmer minimum setpoint or limited hours first) → command acknowledgement → verification → audit. Stop is the last step and is never available for health-sensitive spaces.

- Separate indoor CO₂ ppm from kgCO₂e emissions on every screen.
- Calculate emissions as energy × versioned grid factor; calculate avoided emissions from baseline savings; never call them credits before verification.
- State that a split unit alone recirculates air; fresh-air control needs a connected ventilation device.
- Show data freshness and completeness with every live or aggregated value.

11. Dependencies, assumptions and risks

Item	Type	Handling
Contractual and legal basis for payment-linked restriction per market	Dependency	Legal review in Stage 2 month 1; ladder disabled until sign-off
Infrared codebooks for top brands; VRF and central integration per site	Assumption	Codebook capture in Stage 2 month 1; HVAC priced per site
SDPPI type approval for the RF device	Dependency	Application in month 3; pilot on trial-use nodes
Baseline methodology accepted by a verifier; registry route for AC-efficiency micro-offsets	Risk	Consultant retained; sell Scope 2 reporting first; tokens labelled verified savings certificates
WhatsApp Business API templates approval; payment gateway onboarding	Dependency	Start in Stage 2 month 1
Voice recognition quality in Bahasa Indonesia and other languages	Risk	Text fallback always present; server speech service evaluated in 1B
Gadget cost target at low volumes	Risk	Off-the-shelf modules for pilot; cost-optimised board after pilot

12. Release plan and acceptance

Release	Timing	Contents	Acceptance
R1 Clickable prototype	Stage 1, 2-week sprint	EP-01..EP-15 as simulated experiences; design spec v1; architecture and backend preparation	AC-001..AC-036 of the Product Definition Pack; eight journeys in under 10 minutes
R2 Functional platform	Stage 2 months 1–2	EP-22, EP-06, EP-07, EP-09 live; EP-16, EP-17 initial	Live telemetry from simulator and pilot devices; notifications delivered; payments and ladder working in sandbox

Release	Timing	Contents	Acceptance
R3 Pilot with gadget	Stage 2 months 2–3	EP-20 on 30 units; EP-10, EP-11, EP-12 on metered data	Measured savings vs baseline; tamper events verified; Scope 2 report issued
R4 Hardened launch	Stage 2 month 4	Pen test, SDPPI submission, MRV gap review, mainnet tokens, SLA	Launch checklist; first customers onboarded
R5 HVAC	Phase 2	EP-21	Reference site integrated; HVAC rules validated

13. Open questions

1. Which markets and grids are targeted first after Indonesia, and which languages follow English and Bahasa Indonesia?
2. Commercial model per customer segment: rent-to-own, subscription, maintenance contract, carbon-as-a-service, or mixed?
3. One or two approvers for restriction steps; who may flag a space health-sensitive?
4. Mandatory versus optional sensors on the base gadget variant (CO₂ module as add-on?).
5. Preferred WhatsApp BSP, email, payment gateway and carbon platform per market.
6. Baseline and measurement-and-verification methodology to be proposed to the verifier.
7. Brand, product name and app-store identity for the client app.

14. Notes

- The first deliverable is the clickable UI/UX prototype for split units with simulated data; the gadget and firmware follow in Phase 1C. Sample requirements from phase 1C and 1B is included but will not be the main focus and will not be treated as such, and will be omitted by default in the Phase 1A Prototype in order to meet time constraints.
- Simulations, estimations, and unverified values are to be considered the default in this clickable prototype. Features will be mocked as needed.
- Simulated, estimated and unverified values are to be disclosed when applicable.

15. Traceability summary

Epic	URS families	FR range	Use cases (D1)	Design
EP-01, 02, 13	UR-AUTH, UR-LANG, UR-VOICE	FR-01..07	UC-A01..A05	D2 §3.1, §6
EP-02, 03, 04	UR-ASSET, UR-CLI, UR-TEC, UR-ADM	FR-10..15	UC-B01..B02	D2 §4
EP-05, 06, 16	UR-MNT, UR-COM	FR-20..27	UC-C01..C03	D4 §2–3, D3 §7
EP-07	UR-MNT-07, UR-CLI-09..10, UR-TEC	FR-30..34	UC-E01..E04	D2 §4.3
EP-08, 17	UR-CLI-07, UR-ENE	FR-40..47	UC-D01..D04	D3 §5, D4 §5
EP-09	UR-PAY, UR-CLI-11, UR-ADM-05..07	FR-50..56	UC-F01..F03	D3 §6
EP-10, 11, 18	UR-ENE, UR-AIR	FR-60..66	UC-B03..B04	D4 §4–5
EP-12, 19	UR-CAR	FR-70..76	UC-G01..G04	D3 §8, D4 §6–8
EP-14, 20	UR-DEV	FR-80..86	UC-H01..H02	D3 §3

Epic	URS families	FR range	Use cases (D1)	Design
EP-21	UR-HVAC	FR-90..92	UC-H02 A1	D3 §12
EP-15, 22	UR-ADM, UR-SYS, UR-NFR	FR-100..104	UC-H03..H04	D3 §2, §9–10